



# InventoryOS: Inventory & Warehouse Management System

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## Abstract

InventoryOS is a centralized inventory and warehouse management system that manages products, stock, orders, suppliers, and multiple warehouses. It uses React, Node.js, Express.js, and MySQL to provide real-time inventory updates, smart warehouse allocation, low-stock detection, and stock movement tracking.

## Problem Statement & Objectives

Traditional inventory systems often lack centralized multi-warehouse visibility, real-time stock synchronization, automated reorder detection, and complete traceability of stock movements.

**Objective 1:** [Develop a centralized database for multi-warehouse inventory management] **Objective 2:** [Synchronize customer orders and warehouse inventory through REST APIs] **Objective 3:** [Implement smart warehouse allocation, automated low-stock detection, and stock traceability]

**Scope:** [Users: Customers, Warehouse Managers and Administrators  
Modules: Products, Inventory, Orders, Warehouses, Suppliers, Purchase Orders, Transfers and Stock Movements.]

## Existing vs Proposed System

| Feature                  | Existing System     | Proposed System                     |
|--------------------------|---------------------|-------------------------------------|
| Inventory visibility     | Manual / fragmented | Centralized real-time inventory     |
| Warehouse allocation     | Manual selection    | Automatic warehouse allocation      |
| Low-stock detection      | Manual monitoring   | Automated reorder-level detection   |
| Stock tracking           | Limited history     | Complete stock movement audit trail |
| Multi-warehouse transfer | Manual process      | Digital transfer management         |

## Feasibility Study

- **Technical:** Uses widely supported technologies including React, Node.js, Express.js, MySQL and REST APIs.
- **Economic:** Built using open-source development tools and software, minimizing implementation cost.
- **Operational:** Separate Customer and Warehouse portals provide simple interfaces for ordering and inventory operations.
- **Schedule:** Designed and implemented within the academic semester using an incremental PBL development approach.

## Results & Testing

Multi-warehouse inventory successfully implemented.  
Orders, stock updates, and movements tested.  
Validation and transactions verified.  
Low-stock detection verified.

## Conclusion

InventoryOS provides centralized, synchronized inventory and warehouse management with automated allocation and stock traceability.

## Tools & Technologies

Front-End: React.js, HTML-CSS-JS, Axios / Fetch API

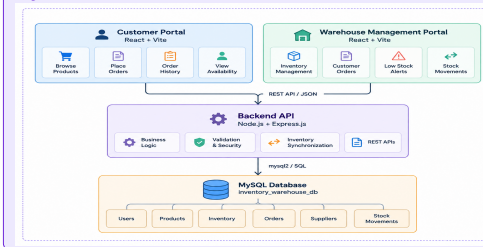
Chart.js

Back-End: Node.js ,Express.js, REST APIs, CORS

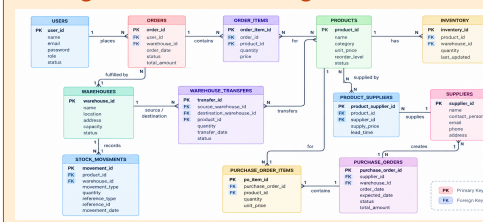
Database: MySQL, mysql2

Tools: Visual Studio Code, MySQL Workbench, Git & GitHub

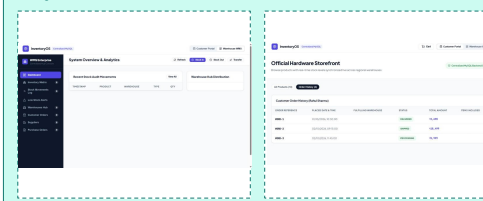
## System Architecture



## ER Diagram & Database Design



## Implementation Screenshots



## Future Scope

- Role-based authentication and authorization
- Advanced inventory analytics and forecasting
- Automated supplier reordering
- Barcode-based stock operations
- Cloud deployment and centralized access